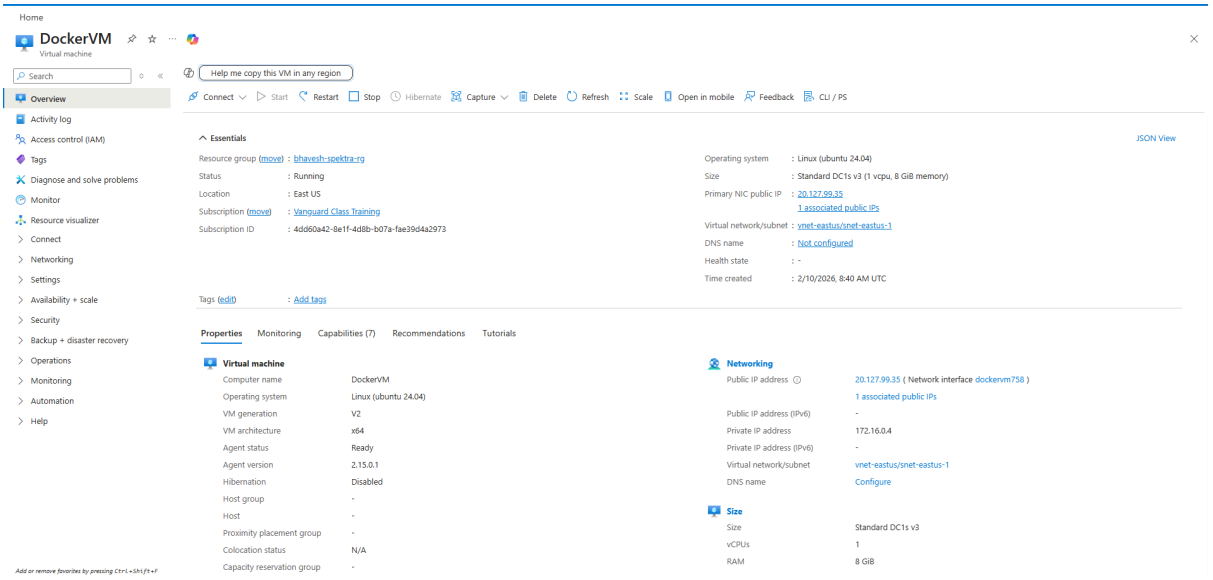
Task 1 - Install Docker and Nginx and host a custom website

Create VM and install Docker on it



Install Docker

```
dockervm@DockerVM:~$ sudo apt install -y docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse
  zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 5 not upgraded.
Need to get 76.1 MB of archives.
After this operation, 288 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 bridge-utils amd64 1.7.1-1ubuntu2 [33.9 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 runc amd64 1.3.3-0ubuntu1~24.04.3 [8815 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 containerd amd64 1.7.28-0ubuntu1~24.04.2 [38.4 MB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 dns-root-data all 2024071801~ubuntu0.24.04.1 [5918 B]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 dnsmasq-base amd64 2.90-2ubuntu0.1 [376 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 docker.io amd64 28.2.2-0ubuntu1~24.04.1 [28.3 MB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 ubuntu-fan all 0.12.16+24.04.1 [34.2 kB]
Fetched 76.1 MB in 5s (15.0 MB/s)
```

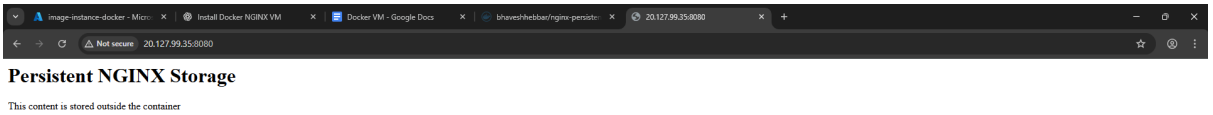
```
dockervm@DockerVM:~$ sudo systemctl start docker
sudo systemctl enable docker
dockervm@DockerVM:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-02-10 08:43:04 UTC; 24s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 3880 (dockerd)
      Tasks: 7
     Memory: 23.5M (peak: 23.8M)
        CPU: 182ms
     CGroup: /system.slice/docker.service
             └─3880 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Feb 10 08:43:03 DockerVM dockerd[3880]: time="2026-02-10T08:43:03.9611143085Z" level=info msg="Loading containers: don
Feb 10 08:43:03 DockerVM dockerd[3880]: time="2026-02-10T08:43:03.985109606Z" level=warning msg="Not using native dif
Feb 10 08:43:03 DockerVM dockerd[3880]: time="2026-02-10T08:43:03.985369953Z" level=info msg="Docker daemon" commit="
Feb 10 08:43:03 DockerVM dockerd[3880]: time="2026-02-10T08:43:03.985510453Z" level=info msg="Initializing buildkit"
Feb 10 08:43:03 DockerVM dockerd[3880]: time="2026-02-10T08:43:03.999929063Z" level=warning msg="CDI setup error /var
Feb 10 08:43:04 DockerVM dockerd[3880]: time="2026-02-10T08:43:04.000276547Z" level=warning msg="CDI setup error /etc
Feb 10 08:43:04 DockerVM dockerd[3880]: time="2026-02-10T08:43:04.024443913Z" level=info msg="Completed buildkit init
Feb 10 08:43:04 DockerVM dockerd[3880]: time="2026-02-10T08:43:04.031808816Z" level=info msg="Daemon has completed in
Feb 10 08:43:04 DockerVM dockerd[3880]: time="2026-02-10T08:43:04.031918022Z" level=info msg="API listen on /run/dock
Feb 10 08:43:04 DockerVM systemd[1]: Started docker.service - Docker Application Container Engine.

dockervm@DockerVM:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
```

```
dockervm@DockerVM:~$ sudo docker run -d   --name nginx-web   -p 8080:80   nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
0c8d55a45c0d: Pull complete
46bf3a120c8e: Pull complete
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
f73400a233fd: Pull complete
47cd406a84ef: Pull complete
bae5a1799a80: Pull complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
e501881ae6df84e6d8c2ea39d025772efbe6ff0c53bb268aad5d5c04004428d9
dockervm@DockerVM:~$ docker ps
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.50/contain
ers/json": dial unix /var/run/docker.sock: connect: permission denied
dockervm@DockerVM:~$ sudo usermod -aG docker $USER
newgrp docker
dockervm@DockerVM:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                                       NAMES
e501881ae6df   nginx     "/docker-entrypoint.…"   22 seconds ago   Up 22 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   nginx-web
```

Website uploaded through nginx using VM's Public IP



Task 2 - Create Persistent Storage

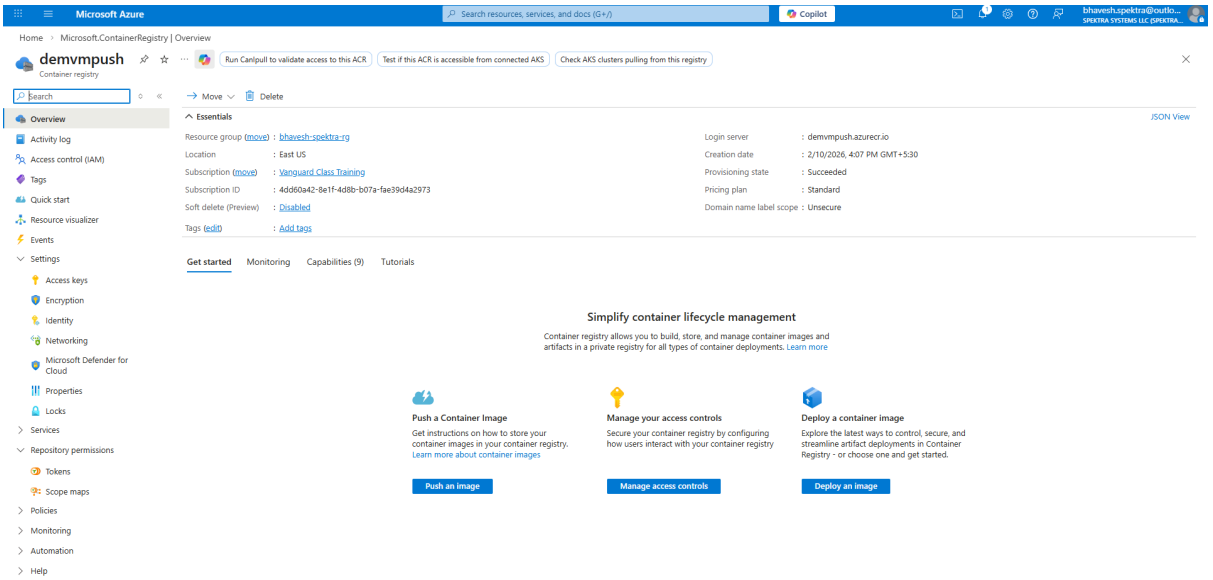
Persistent Storage

```
dockervm@DockerVM:~$ sudo mkdir -p data/nginx/html
dockervm@DockerVM:~$ sudo chown -R $user:$user /data/nginx
chown: cannot access '/data/nginx': No such file or directory
dockervm@DockerVM:~$
sudo chown -R $USER:$USER /data/nginx
chown: cannot access '/data/nginx': No such file or directory
dockervm@DockerVM:~$ sudo mkdir -p data/nginx/html
dockervm@DockerVM:~$ cat data/nginx/html
cat: data/nginx/html: Is a directory
dockervm@DockerVM:~$ sudo mkdir -p data/nginx/html
dockervm@DockerVM:~$ sudo chown -R $USER:$USER /data/nginx
chown: cannot access '/data/nginx': No such file or directory
dockervm@DockerVM:~$ cd ...
bash: cd: ...: No such file or directory
dockervm@DockerVM:~$ cd ..
dockervm@DockerVM:/home$ sudo mkdir -p data/nginx/html
dockervm@DockerVM:/home$ sudo chown -R $USER:$USER /data/nginx
chown: cannot access '/data/nginx': No such file or directory
dockervm@DockerVM:/home$ sudo chown -R $USER:$USER /data/nginx
chown: cannot access '/data/nginx': No such file or directory
dockervm@DockerVM:/home$ sudo mkdir -p /data/nginx/html
dockervm@DockerVM:/home$ sudo chown -R $USER:$USER /data
dockervm@DockerVM:/home$ ls -ld /data /data/nginx /data/nginx/html
drwxr-xr-x 3 dockervm dockervm 4096 Feb 10 08:59 /data
drwxr-xr-x 3 dockervm dockervm 4096 Feb 10 08:59 /data/nginx
```

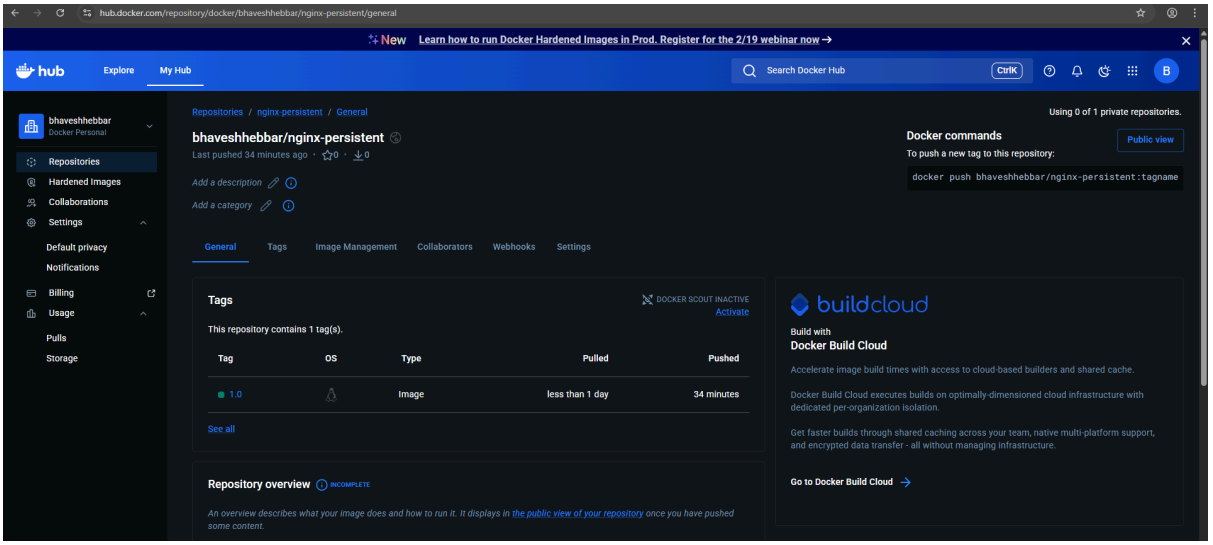
```
dockervm@DockerVM:~/nginx-image$
dockervm@DockerVM:~/nginx-image$ docker ps -a
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS
                 NAMES
ef47ca8b3da1   nginx-persistent:1.0   "/docker-entrypoint.…"   51 minutes ago   Up 51 minutes   0.0.0.0:8080->80/tcp, [:
:]:8080->80/tcp   nginx-test
dockervm@DockerVM:~/nginx-image$ docker rm -f ef47
ef47
dockervm@DockerVM:~/nginx-image$ docker run -d \
  --name nginx-persistent \
  -p 8080:80 \
  -v /data/nginx/html:/usr/share/nginx/html \
  nginx
e1bd70c1db1daa85ba26ffc06d7ca57d133f1803a78f087bd312fdc69cc6a982
dockervm@DockerVM:~/nginx-image$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS
   NAMES
e1bd70c1db1d   nginx     "/docker-entrypoint.…"   6 seconds ago   Up 5 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
   nginx-persistent
```

Task 3 - Create Container Registry

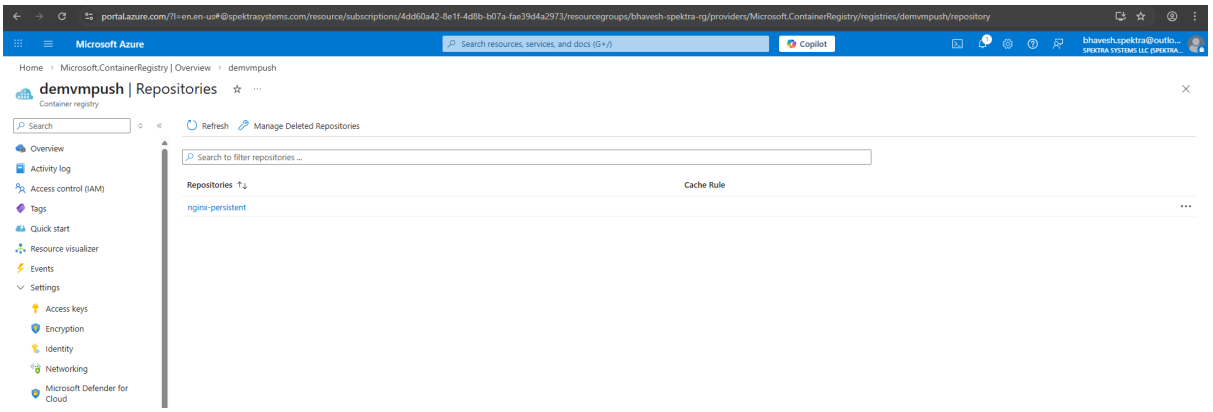
ACR



Docker image pushed to docker hub

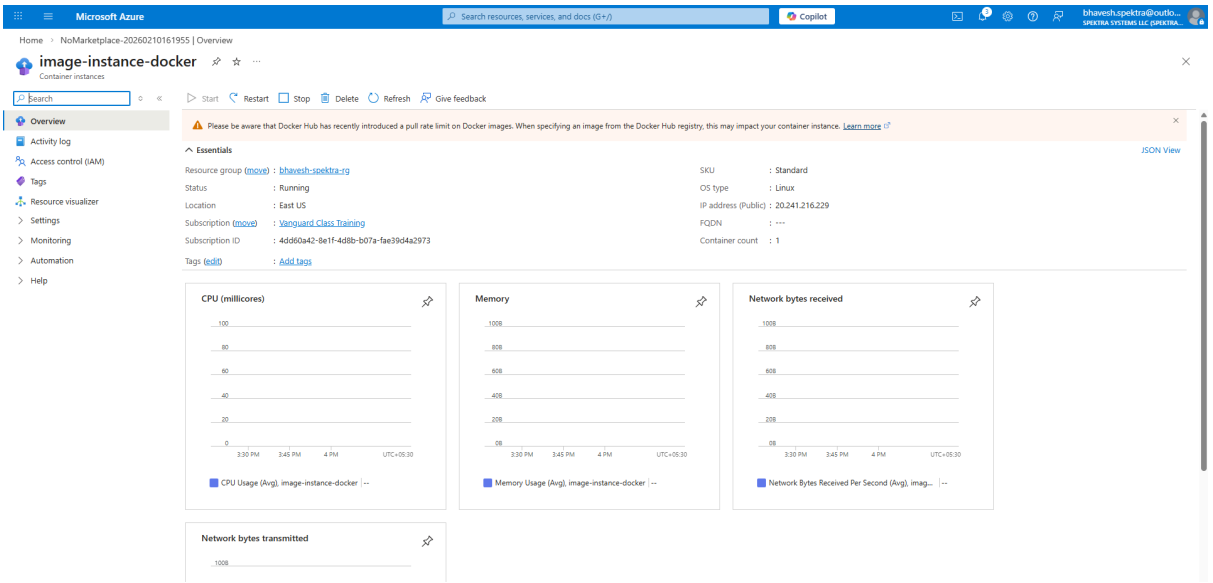


ACR repo created - image pushed from docker hub to ACR

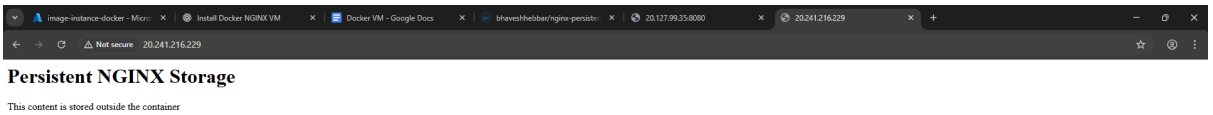


Task 4 - Run instance through Container Instances

Container Instance created -



Instance accessed through Public IP



Persistent NGINX Storage

This content is stored outside the container